

PROFESSIONAL EXPERIENCE

- Oct 2022 – **Software Development Engineer** ♦ Amazon Web Services, Seattle, WA
 Jan 2025 *Python, Pytest, Flask, Ruby, Ruby-on-Rails, bash scripting, software deployment, various AWS services*
- Created distributed agent system to upkeep 240+ EC2 instances and across 40+ regions under OS patching campaign, providing an automated solution to meet security SLAs
 - Developed dashboard plot feature for diagnostics website to enhance visibility in NLB network traffic, resulting in faster access to metrics for troubleshooting
 - Automated logic to bulk refresh ALB/CLB API workflows as batch jobs of up to 500 workflows at a time, leading to large time savings for engineers when retrying failed workflows on diagnostics website
 - Delivered infrastructure build for canary testing service and diagnostics website to 9 new regions, ensuring new AWS regions receive testing of different load balancer services on schedule

EDUCATION

- Jan 2025 – **GEORGIA INSTITUTE OF TECHNOLOGY** ♦ ATLANTA, GA
 Dec 2026 **Degree:** Computer Science M.S. **GPA:** 4.00 / 4.00
Courses: *Software Analysis, Network Science, Advanced Operating Systems, Deterministic Optimization*
- Aug 2019 – **UNIVERSITY OF FLORIDA** ♦ GAINESVILLE, FL
 May 2022 **Degree:** Computer Science B.S. & Statistics B.A. **GPA:** 3.76 / 4.00
Courses: *Operating Systems, Machine Learning, Statistical Learning, Competitive Prog., Data Science*

RESEARCH & PROJECT EXPERIENCE

- Jan 2022 – **Network Cache System Simulator** ♦ Probability in CS course, UF
 Apr 2022 *Julia, simulation study, caching*
- Designed caching policy for discrete-event simulation study of a network file system, which was tested against well-known caching policies in literature
 - Ranking based on average network request completion time, performance analysis revealed that evicting largest file first is fastest on average
- Jun 2021 – **Data Analytics and Info Retrieval at Univ. of North Texas** ♦ NSF REU, Dept. of Information Science, UNT
 Aug 2021 *Python, information retrieval, transfer learning, deep learning*
Publication: <https://www.naturl.link/unt-reu>
- Experimentally evaluated traditional versus deep learning algorithms on legal text classification to identify models that could help legal teams quickly identify legal contract type from corporate documents
 - Analyzed dataset factors to explain why traditional ML, especially boosting algorithms, outperformed deep learning models transferred from BERT and Sentence-BERT models
- May 2020 – **Predictive Sampling Method for Spread Models in Networks** ♦ University Scholars Program, UF
 Apr 2021 *Python, network science, simulation study*
Publication: <https://www.naturl.link/uf-jur>
- Developed a novel, trainable sampling method for large networks by targeting high-degree nodes in a network to test the addition of parameter tuning that is largely lacking in existing sampling algorithms
 - Applied statistical measurements to compare with existing methods to crawl networks, which hinted community detection applications for the new algorithm particularly in dense networks
- May 2020 – **Backend Engineer for StudioOCR Note-taking Application** ♦ Performant Programming course, UF
 Aug 2020 *Python, OpenCV*
Pip package: <https://pypi.org/project/StudioOCR/>
- Built machine learning and database integration for optical character recognition (OCR) desktop application, which allows users to convert handwritten notes to digitally typed format
 - Engineered pipeline to pre-process text images with OpenCV and Pillow, setting up an experimental unit to tune parameters and test image filters before applying OCR to boost text prediction accuracy

PROFESSIONAL SKILLS

Programming and Technical Languages

[Python](#), [Ruby](#), [JavaScript/Typescript](#), [C++](#), [SQL](#), [HTML/CSS](#), [XML](#), [JSON](#), [YAML](#)

Software Tools, Frameworks, and Services

[AWS EC2](#), [VPC](#), [Lambda](#), [DynamoDB](#), [RDS](#), [SSM](#), [ECS](#), [SQS](#), [R53](#), [Cloudformation](#), [CloudWatch](#)

[Ruby-on-Rails](#), [Flask](#), [Linux utilities](#), [bash scripting](#), [Docker](#), [Kubernetes](#), [PostgreSQL](#), [Node.js](#)